Computer Networks
Reference Models

Prepared by
Dr. Suvarna Sharma (CSE)

Department of Computer Science and Applications,
MITWPU, Pune

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Protocols and Models

- **How a network device operates within a network?**

A network device is a node in the wireless mesh network. It can transmit and receive wireless data and perform the basic functions necessary to support network formation and maintenance.

Communications Fundamentals - Networks vary in size, shape, and function. For communication to occur, devices must know "how" to communicate, three elements in common:

- **Message source (sender)** - Message sources are people, or electronic devices, that need to send a message to other individuals or devices.
- **Message Destination (receiver)** - The destination receives the message and interprets it.
- **Channel** - This consists of the media that provides the pathway over which the message travels from source to destination.

Sending a message, whether by face-to-face communication or over a network, is governed by rules called protocols. Protocols must account for the following requirements to successfully deliver a message that is understood by the receiver: An identified sender and receiver, Common language and grammar, Speed and timing of delivery, Confirmation or acknowledgment requirements.

Protocols and Models

Network Protocol Requirements - Common computer protocols include the following requirements:

- Message encoding
- Message formatting and encapsulation
- Message size
- Message timing
- Message delivery options

Network Protocol Overview

Protocols are implemented by end devices and intermediary devices in software, hardware, or both. Each network protocol has its function, format, and rules for communications.

The table lists the various types of protocols that are needed to enable communications across one or more networks.

Protocol Type	Description
Network Communications Protocols	Protocols enable two or more devices to communicate over one or more networks. The Ethernet family of technologies involves a variety of protocols such as IP, Transmission Control Protocol (TCP), HyperText Transfer Protocol (HTTP), and many more.
Network Security Protocols	Protocols secure data to provide authentication, data integrity, and data encryption. Examples of secure protocols include Secure Shell (SSH), Secure Sockets Layer (SSL), and Transport Layer Security (TLS).
Routing Protocols	Protocols enable routers to exchange route information, compare path information, and then to select the best path to the destination network. Examples of routing protocols include Open Shortest Path First (OSPF) and Border Gateway Protocol (BGP).
Service Discovery Protocols	Protocols are used for the automatic detection of devices or services. Examples of service discovery protocols include Dynamic Host Configuration Protocol (DHCP) which discovers services for IP address allocation, and Domain Name System (DNS) which is used to perform name-to-IP address translation.

The functions of these protocols are addressing, reliability, flow control, sequencing, error detection, and application interface.

OSI Reference Model

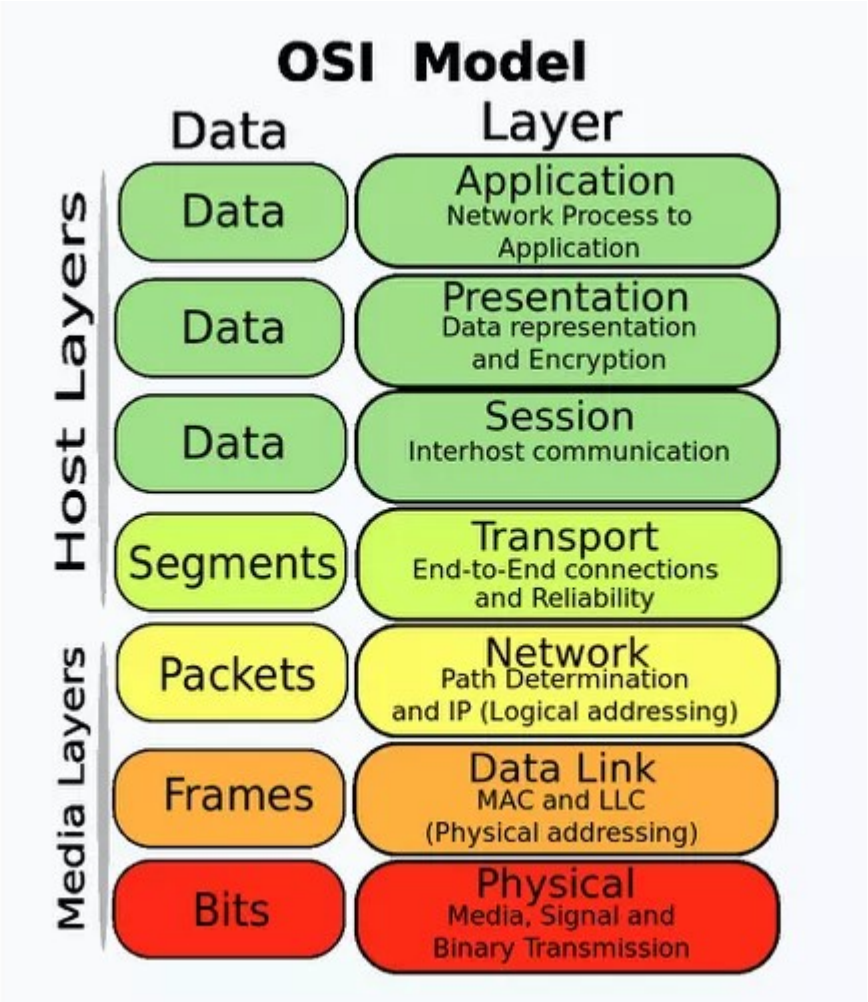

Figure 1 OSI Reference Model

Benefit of using layered model

These are the benefits of using a layered model to describe network protocols and operations as shown in Figure 3:

- Assisting in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below
- Fostering competition because products from different vendors can work together
- Preventing technology or capability changes in one layer from affecting other layers above and below
- Providing a common language to describe networking functions and capabilities
- Two layered models used: Open System Interconnection (OSI) Reference Model and TCP/IP Reference Model

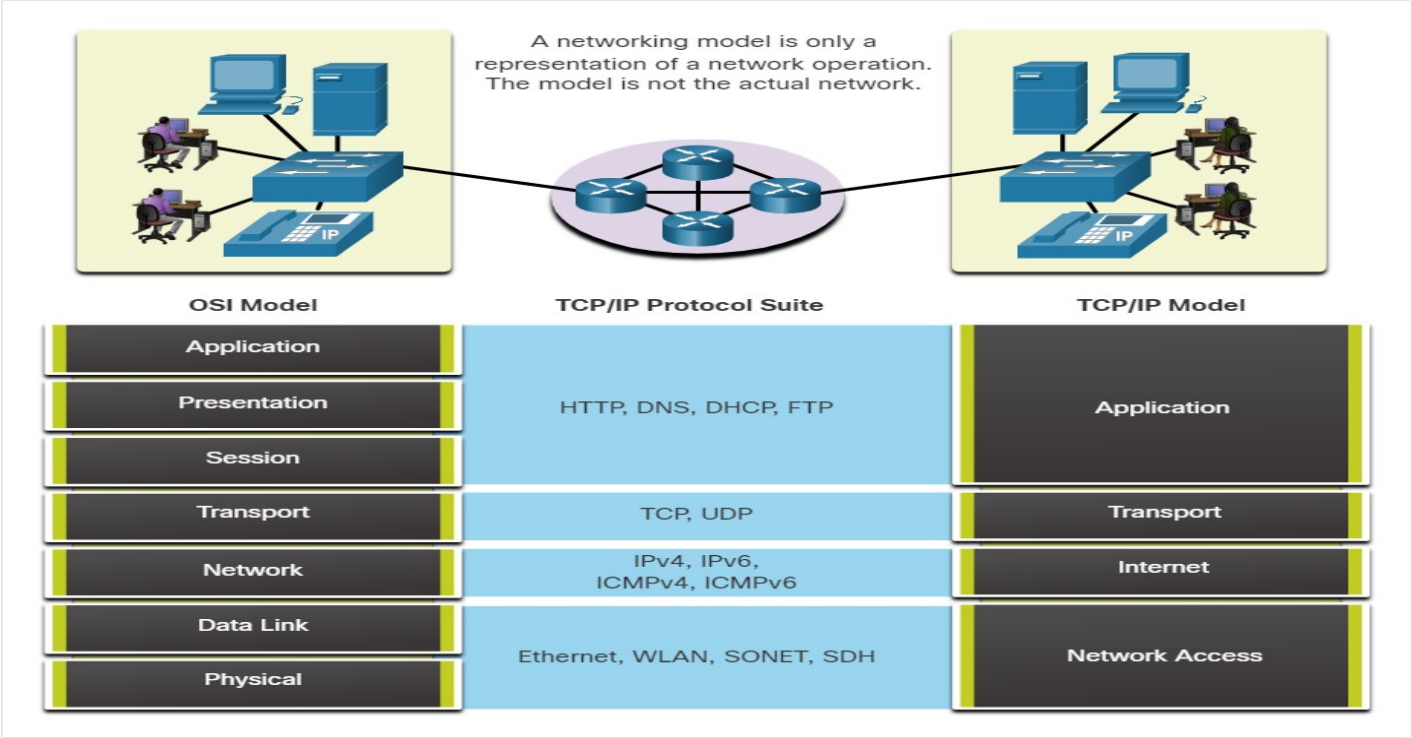

Figure 2 Shows network protocols and operations of layered model

OSI Reference Model

- OSI Reference Model - **Internationally Standardized Network Architecture.**
- OSI = **Open Systems Interconnection**: deals with open systems, i.e. **systems open for communications with other systems**.
- Specified in **ISO 7498**.
- Model has **7 layers**.

7-Layer OSI Model

Figure 3 displays 7 layers of OSI model and overview of their specific task.

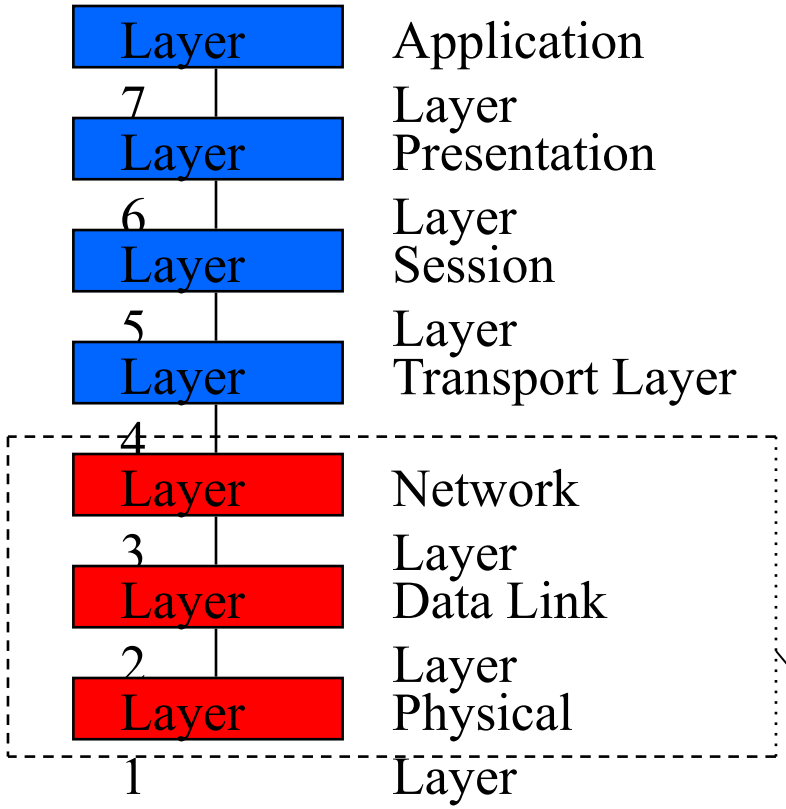

Figure 3 Shows OSI model layers

- Layers 1-4 relate to communications technology.
- Layers 5-7 relate to user applications.

Communications subnet boundary

Reference model (OSI)

- OSI: Open System Interconnect
- Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

Note

ISO is the organization.
OSI is the model.

Layers of OSI model

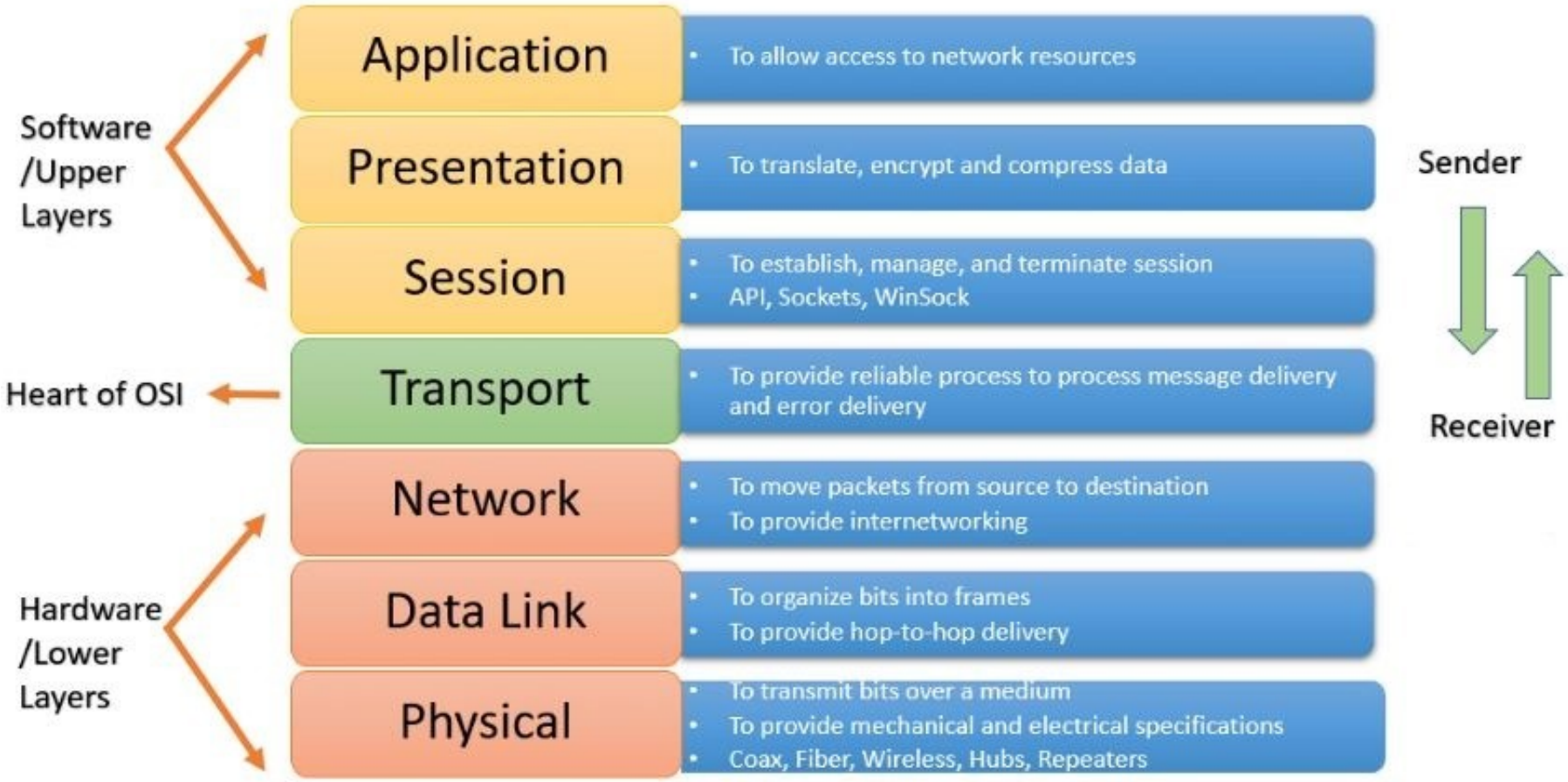

Figure 4 Shows responsibilities of OSI model layers

Interaction between layers in OSI model

Figure 5 shows interaction between layers in OSI model at each layer in detail.

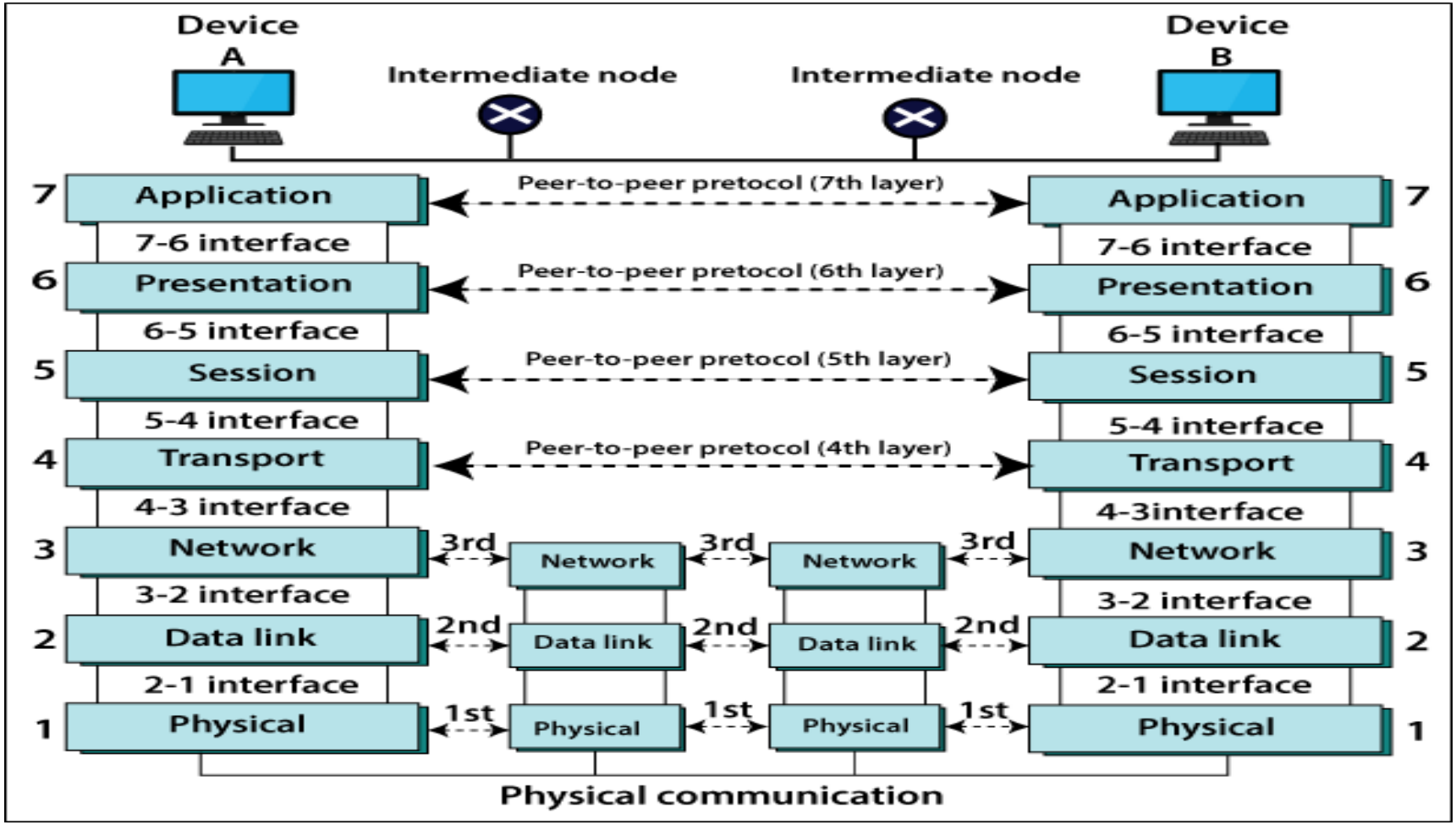

Figure 5 Shows interaction between layers in OSI model.

An exchange using the OSI model

Figure 6 shows the exchange of information using OSI model and transmission medium.

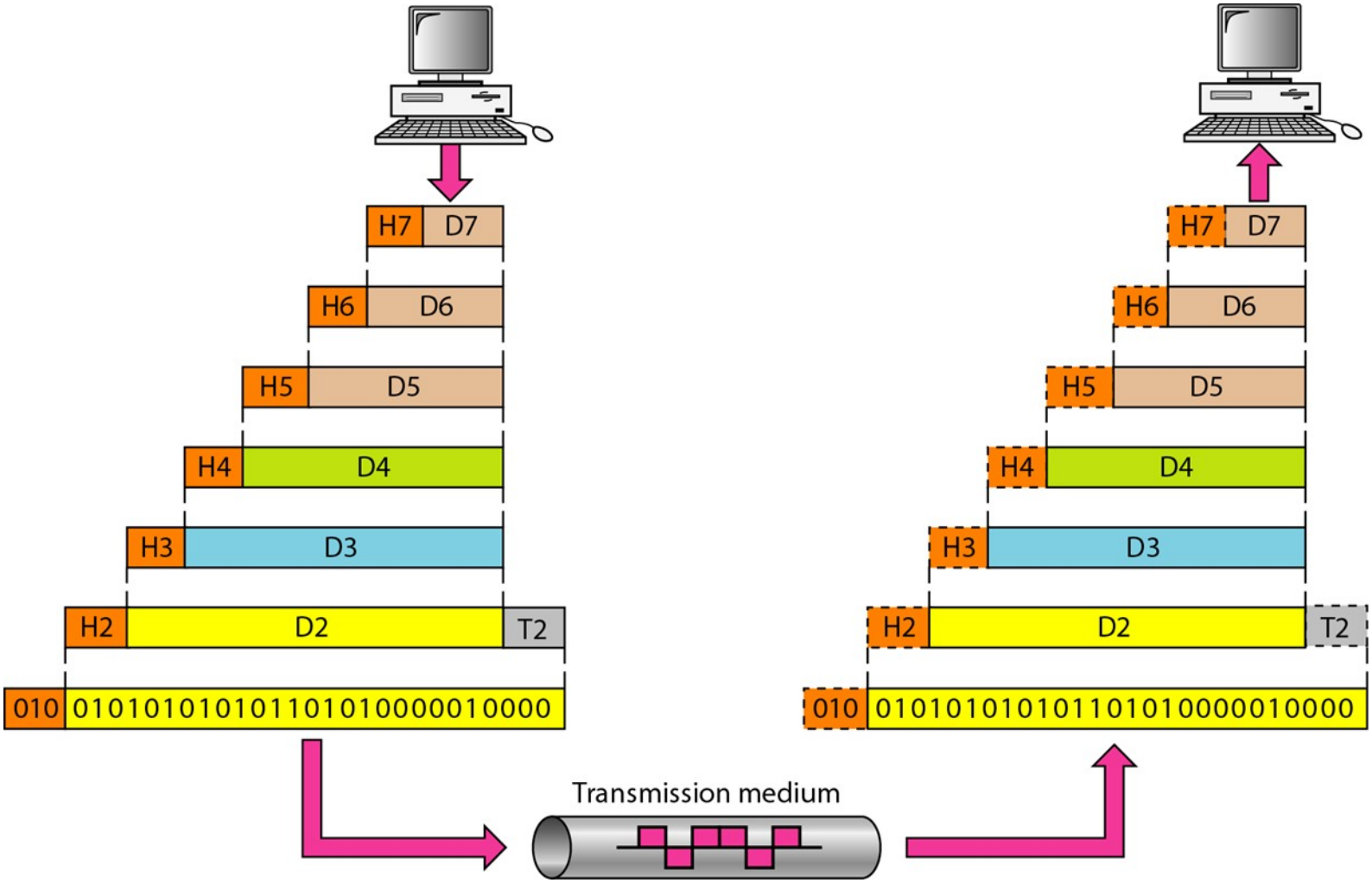

Figure 6 Exchange of information using OSI model.

Layers in the OSI model

- In this section we briefly describe the functions of each layer in the OSI model.
  - Physical Layer
  - Data Link Layer
  - Network Layer
  - Transport Layer
  - Session Layer
  - Presentation Layer
  - Application Layer

Layer 1 :Physical Layer

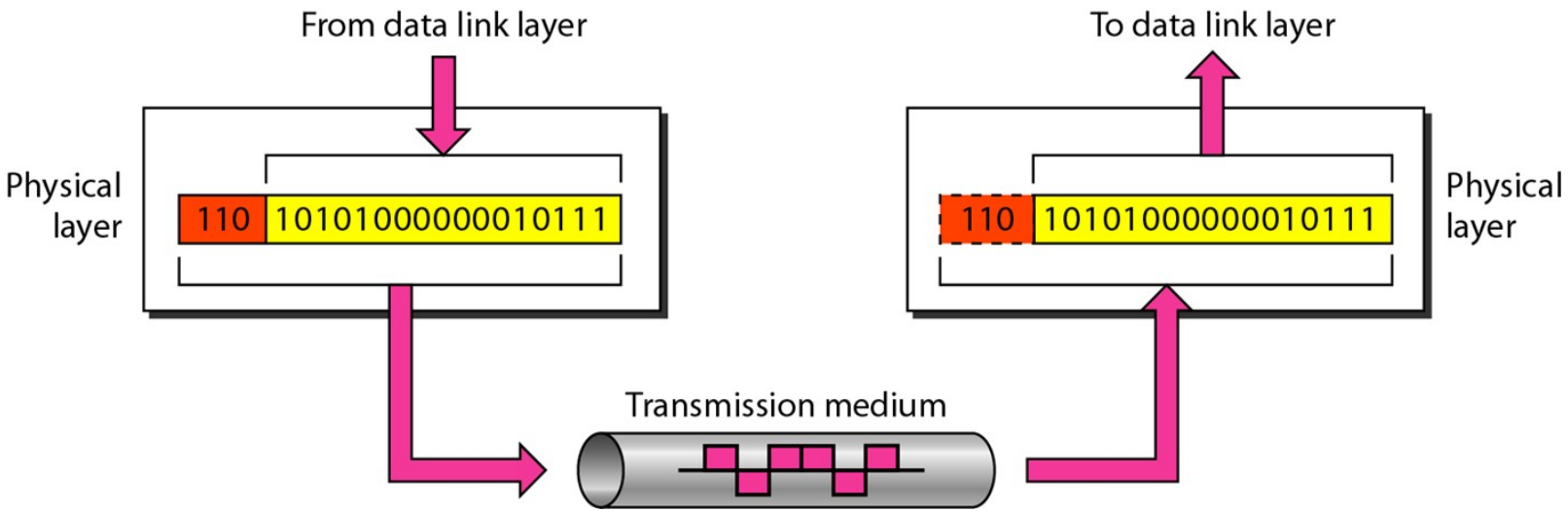

Figure 7 Shows the physical layer information transmitted in OSI model.

Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Layer 2 : Data link layer

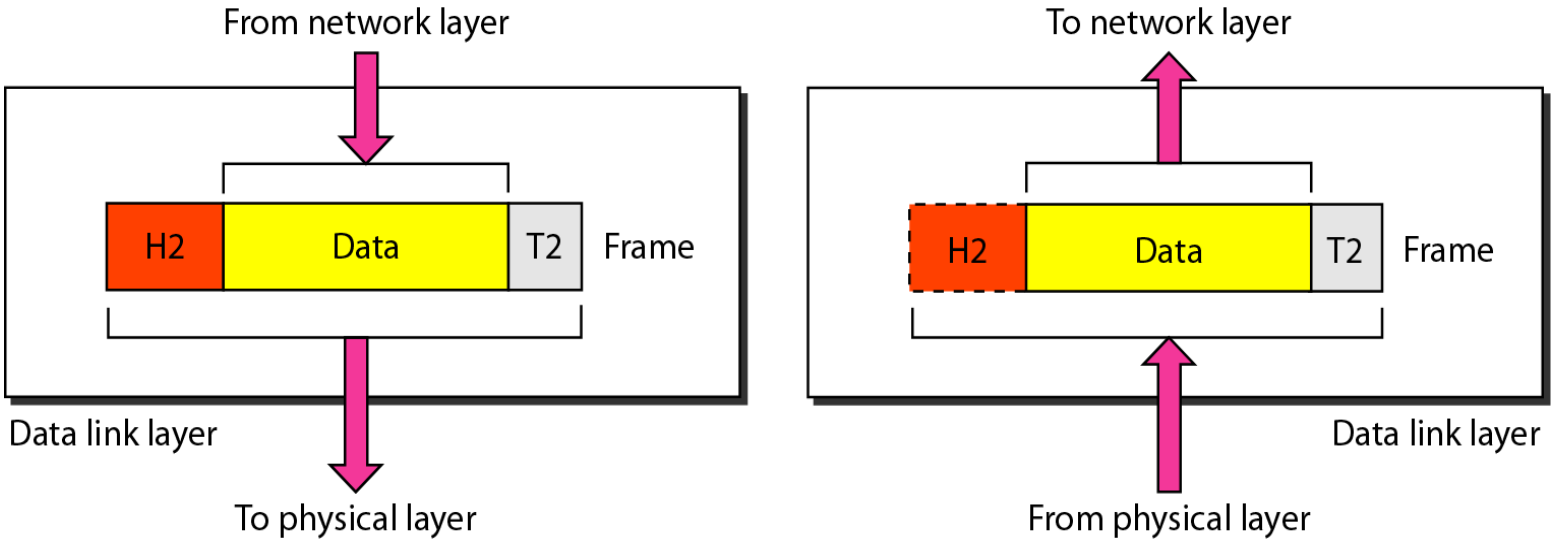

Figure 8 Shows the data link layer information transmitted in OSI model.

Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Layer 3 : Network layer

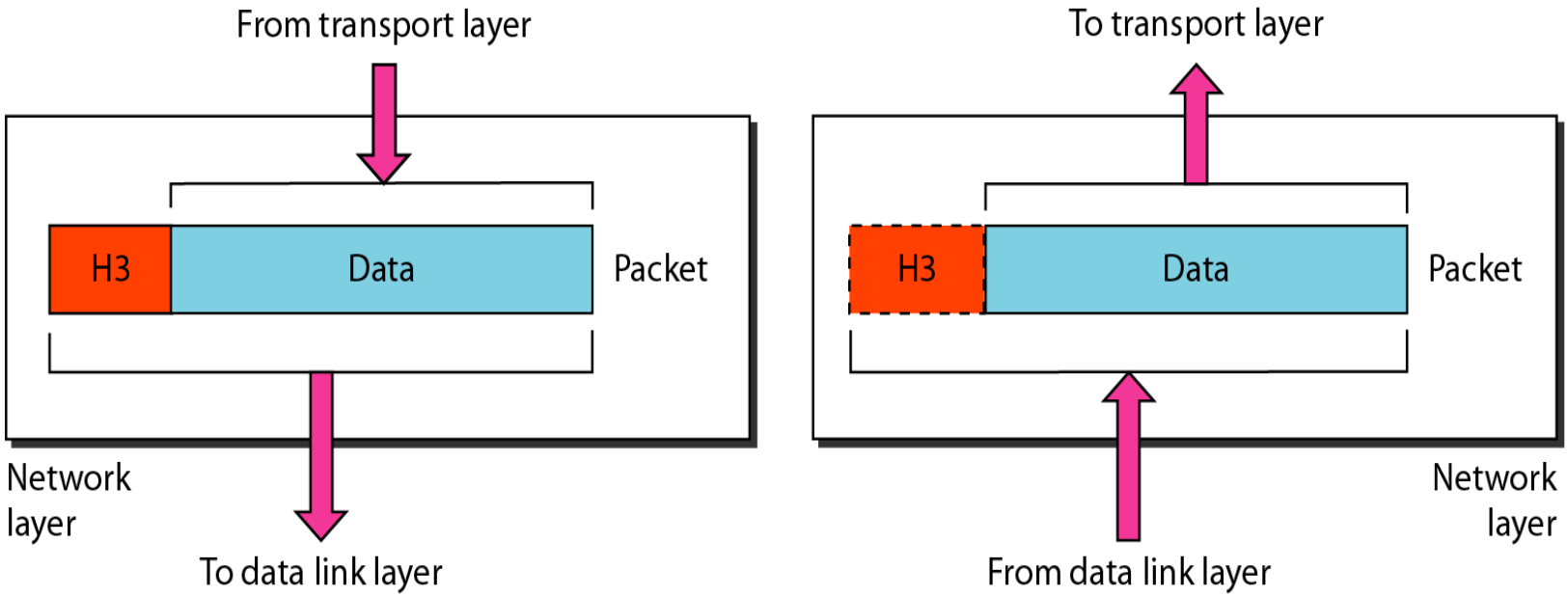

Figure 9 Shows the network layer information transmitted in OSI model.

Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Source to destination delivery

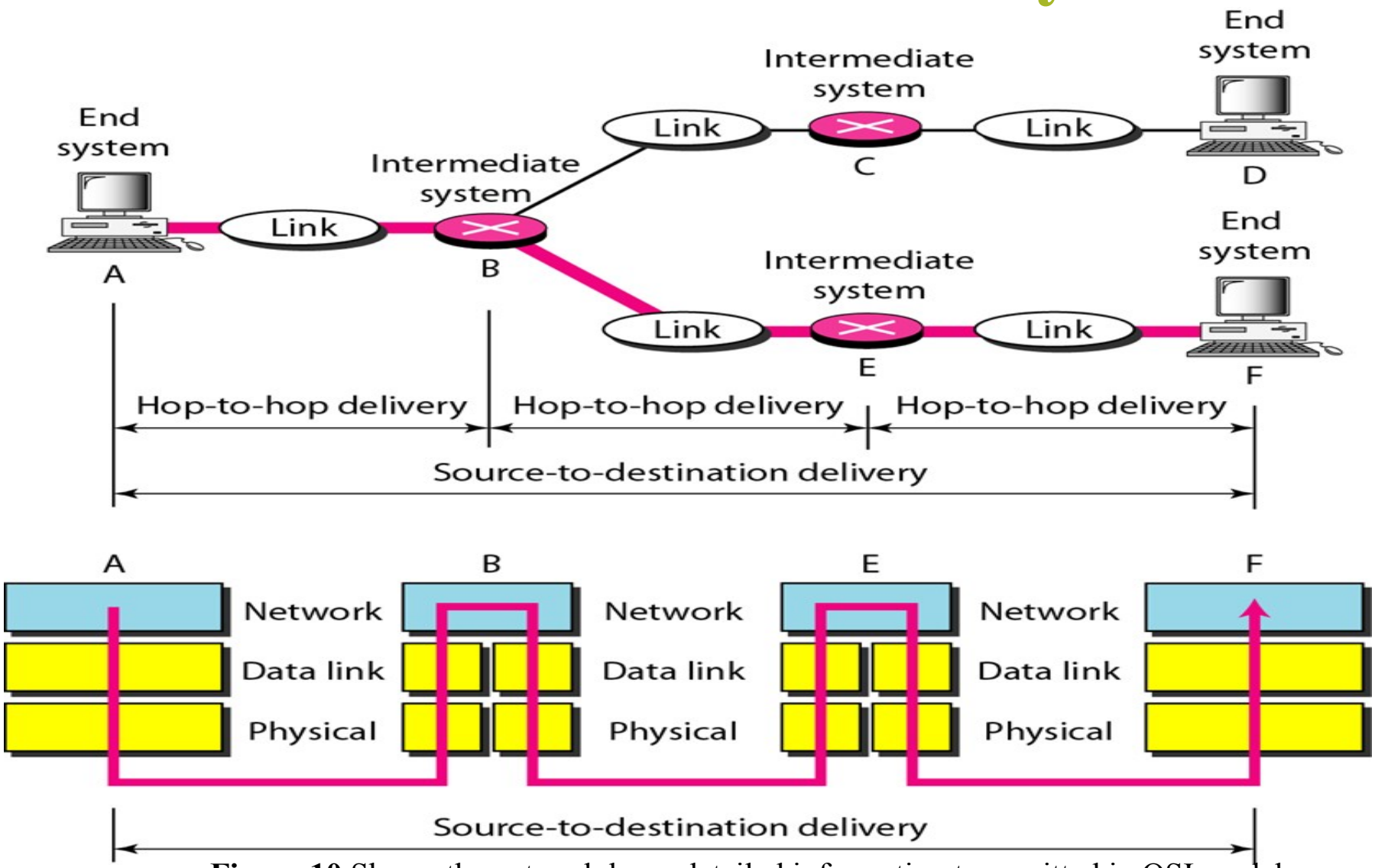

Figure 10 Shows the network layer detailed information transmitted in OSI model.

Layer 4: Transport Layer

- Manages transmission packets
  1. Repackages long messages when necessary, into small packets for transmission
  2. Reassemble packets in the correct order to get the original message.
- Handles error recognition and recovery.
  1. Transport layer at receiving acknowledges packet delivery.
  2. Resends missing packets

- Transport Layer is responsible for end-to-end connectivity. It is also known as the heart of OSI Layers. Following tasks are performed at the Transport Layer : -
  - Identifying Service
  - Multiplexing & De-multiplexing
  - Segmentation
  - Sequencing & Reassembling
  - Error Correction
  - Flow Control
  - Transport

Layer 4: Transport Layer

- **Identifying Service:** Uses port numbers (e.g., port 80 for HTTP) to direct data to the correct application process on the destination host, enabling multiple applications to use the network simultaneously (multiplexing).
- **Multiplexing & De-multiplexing:** Multiplexing combines data from multiple application processes into one stream for the network, while de-multiplexing separates incoming data back to the correct application on the receiving end, using port numbers.
- **Segmentation:** Breaks large chunks of data from the Session Layer into smaller, manageable segments for network transmission and reassembles them in the correct order at the destination.
- Sequencing & Reassembling:
- **Error Control:** Ensures reliable delivery through acknowledgments and retransmissions if data is lost or corrupted.
- **Flow Control:** Prevents a fast sender from overwhelming a slow receiver.
- **Transport:** Establishes, maintains, and terminates connections (e.g., TCP) or provides connectionless services (e.g., UDP).

Transport layer

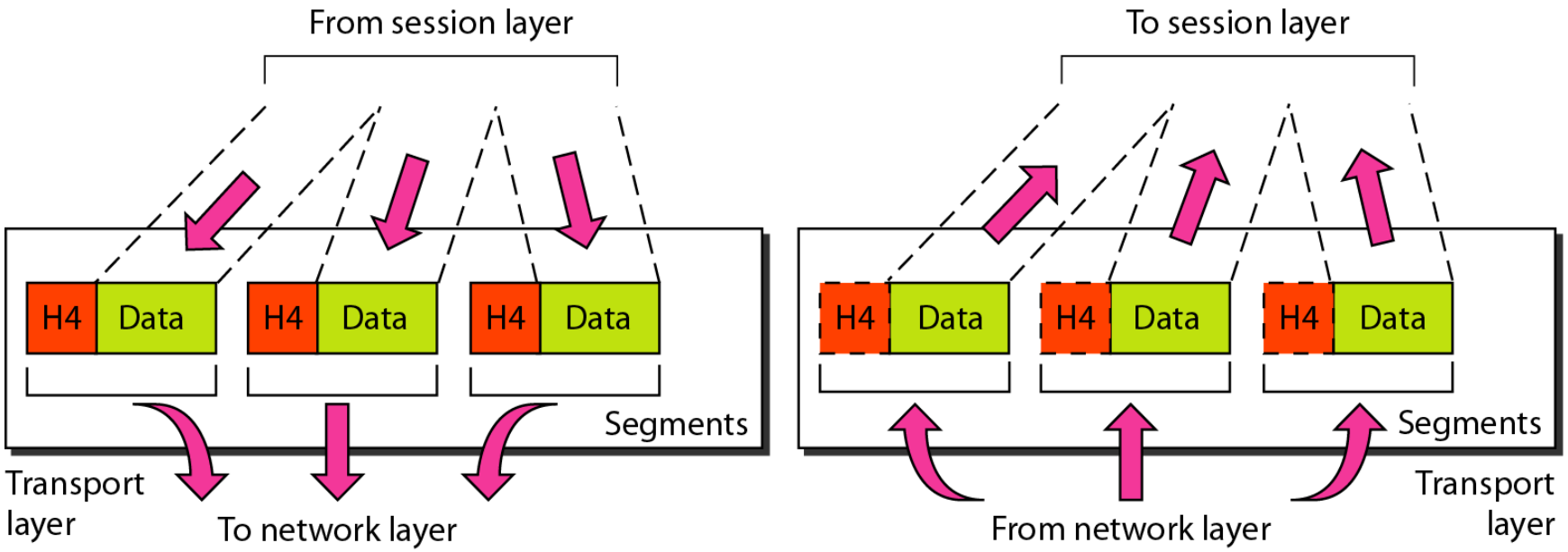

Figure 11 Shows the transport layer information transmitted in OSI model.

Note

The transport layer is responsible for the delivery of a message from one process to another.

TCP	UDP
• Transmission Control Protocol	• User Datagram Protocol
• Connection Oriented	• Connection Less
• Acknowledgement	• No Acknowledgement
• Reliable	• Unreliable
• Slower	• Faster
• Port No. 6	• Port No. 17
• e.g. HTTP, FTP, SMTP	• e.g. DNS, DHCP, TFTP

Figure 12 TCP Vs UDP

Segmentation

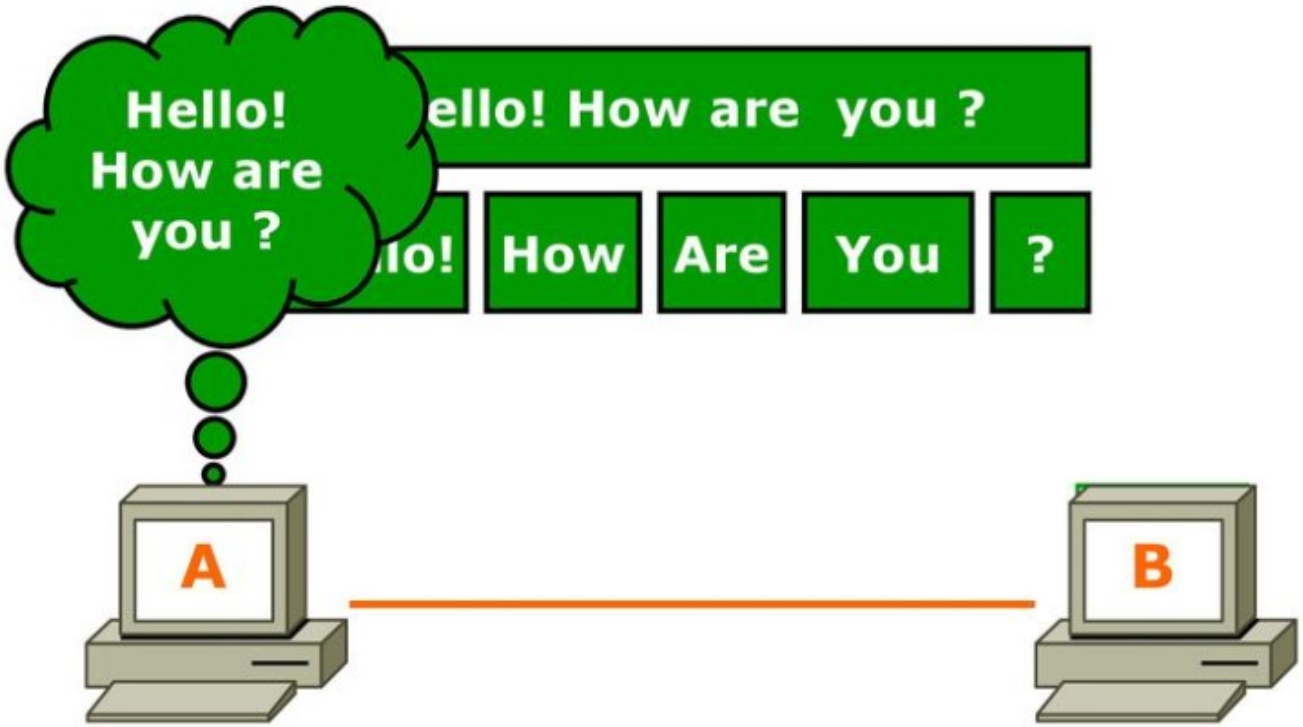

Figure 13 Segmentation process

Sequence & Reassembling

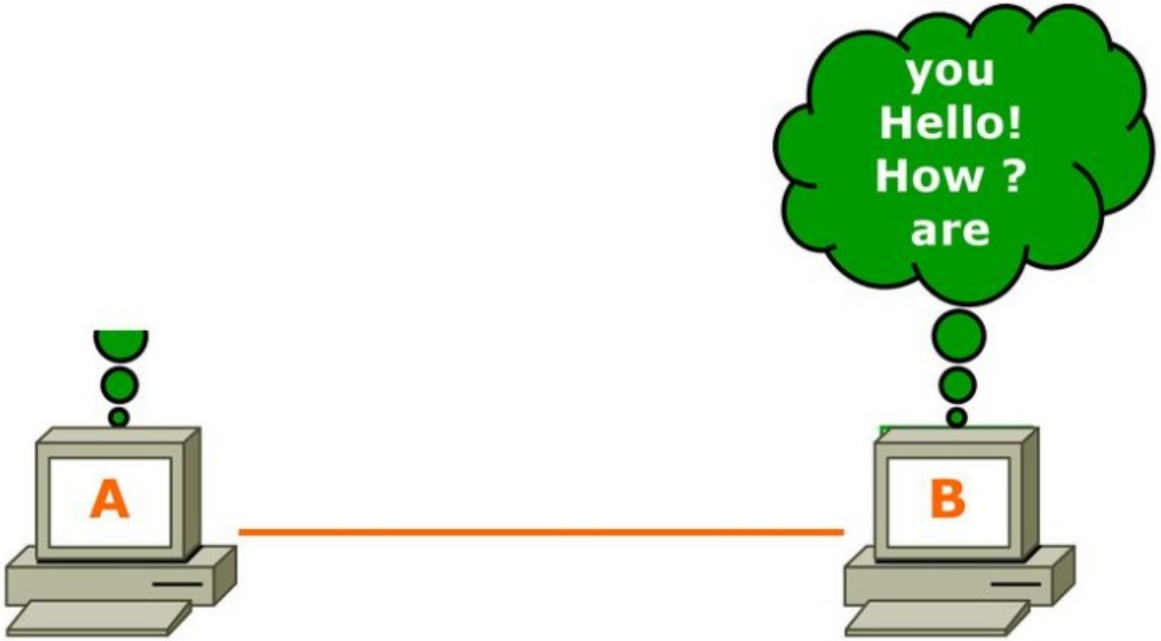

Figure 14 Sequence & Reassembly

Layer 5: Session Layer

- Allows two applications on different computers to establish, use, and end a session. e.g. file transfer, remote login
- Establishes dialog control -Regulates which side transmits, plus when and how long it transmits.
- Performs token management and synchronization.
- Session Layer is responsible for establishing, maintaining and terminating session. Session ID works at Session Layer.

Examples :

- RPC - Remote Procedure Call
- SQL - Structured Query Language
- NFS - Network File System

Session layer

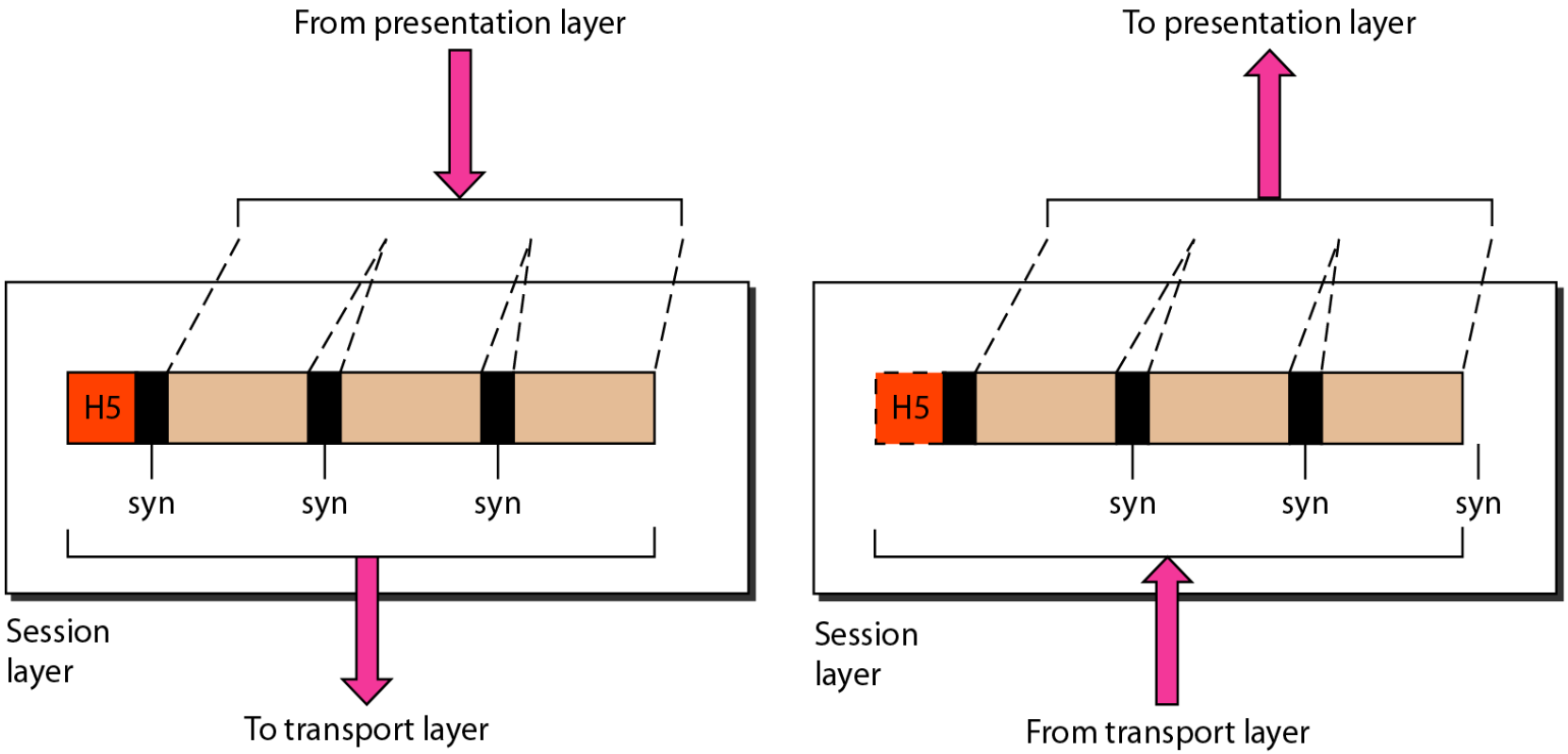

Figure 15 Shows the session layer information transmitted in OSI model.

Note

The session layer is responsible for dialog control and synchronization.

Layer 6 -Presentation layer

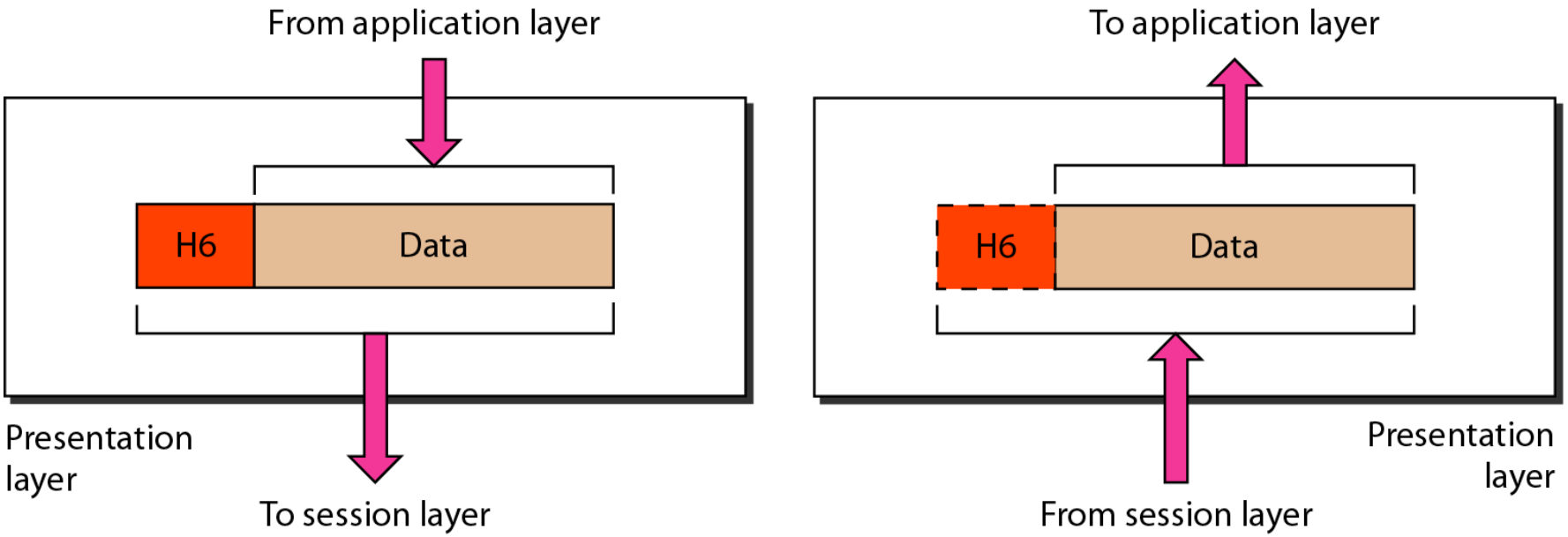

Figure 16 Shows the presentation layer information transmitted in OSI model.

Note

The presentation layer is responsible for translation, compression, and encryption.

Layer 7 - Application layer

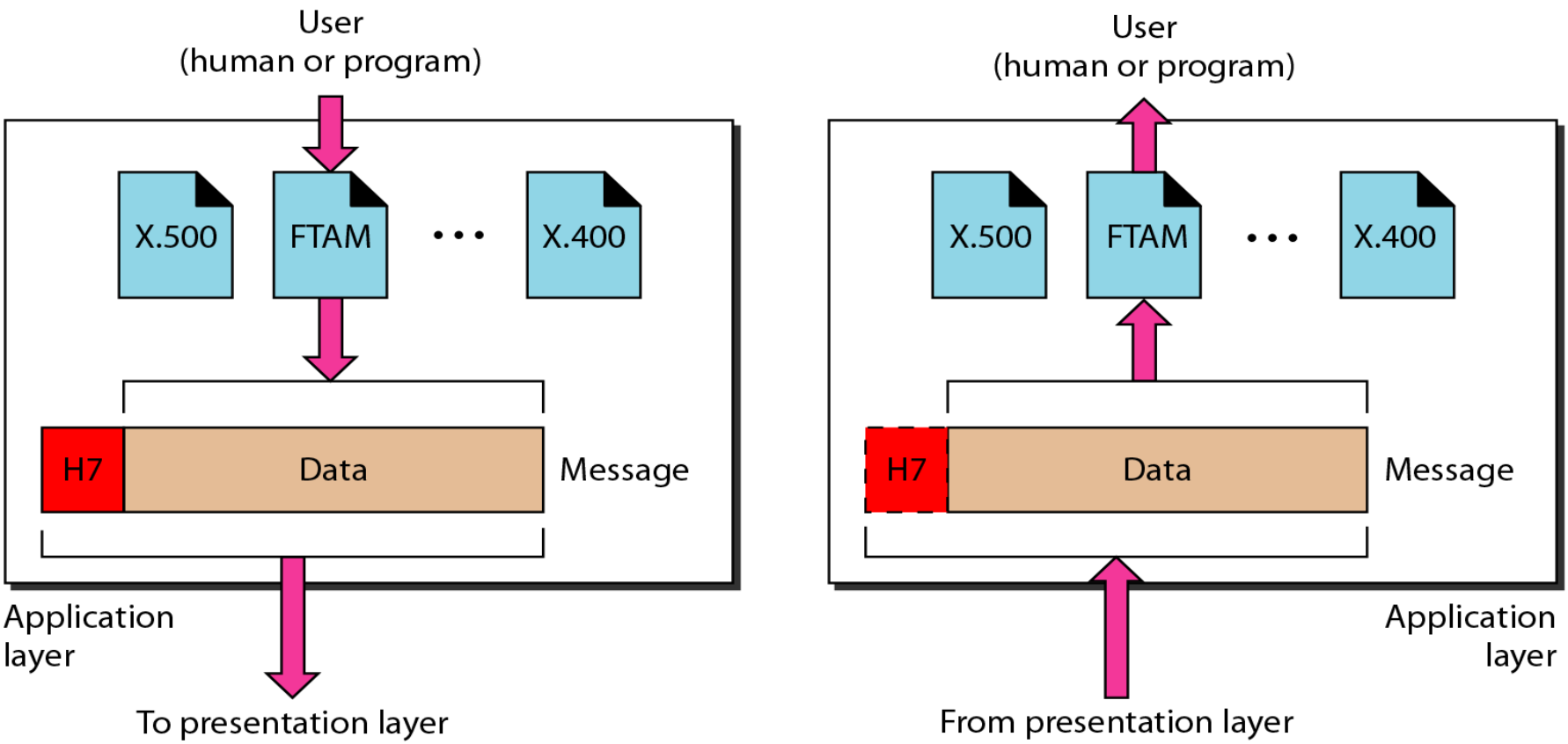

Figure 17 Shows the application layer information transmitted in OSI model.

Note

The application layer is responsible for providing services to the user.

Summary of layers

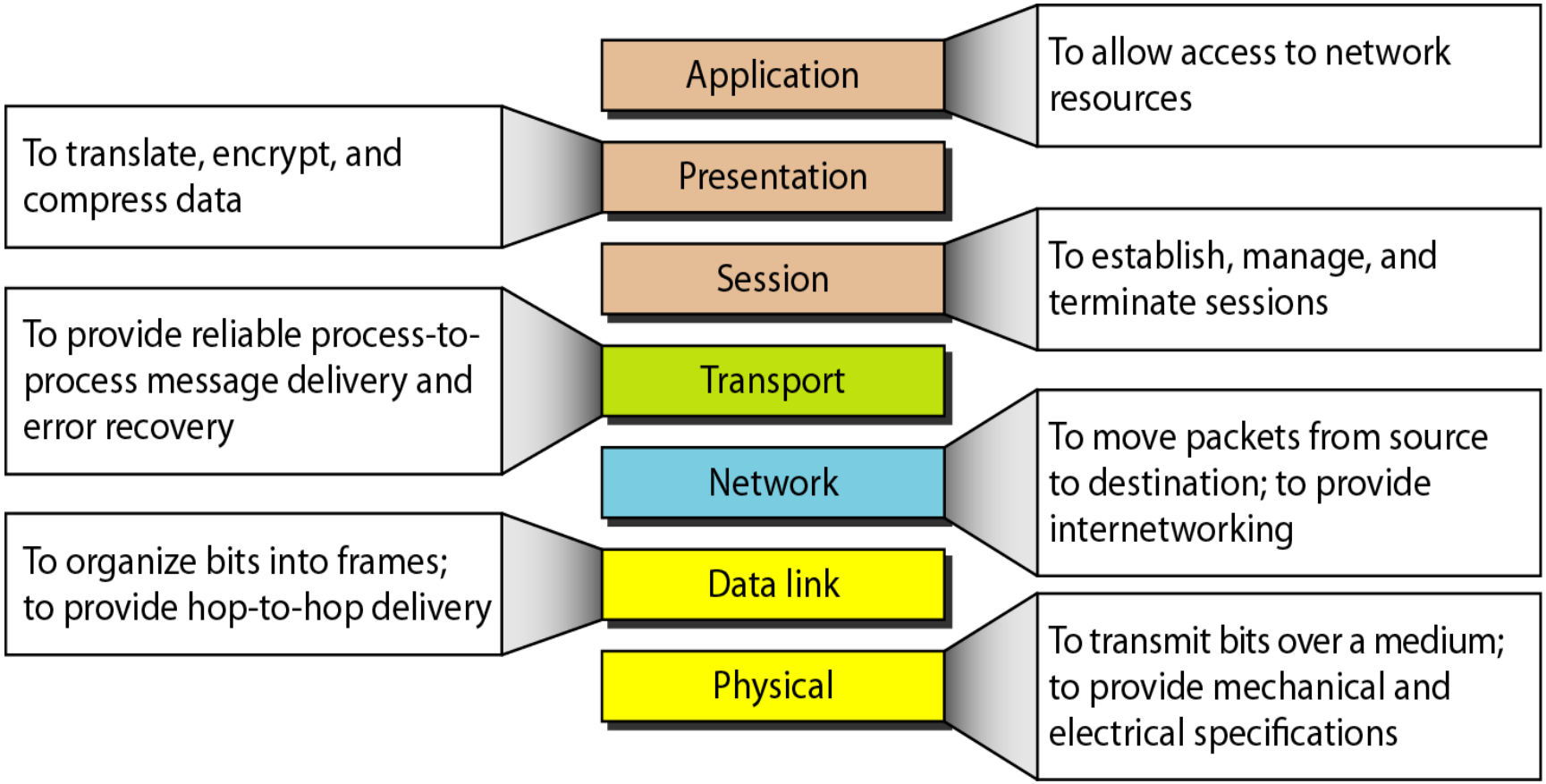

Figure 18 Presents the summary of layers of OSI model.

Protocol supported at various layers

Layer	Name	Protocols
Layer 7	Application	SMTP, HTTP, FTP, POP3, SNMP
Layer 6	Presentation	MPEG, ASCH, SSL, TLS
Layer 5	Session	NetBIOS, SAP
Layer 4	Transport	TCP, UDP
Layer 3	Network	IPV5, IPV6, ICMP, IPSEC, ARP, MPLS.
Layer 2	Data Link	RAPA, PPP, Frame Relay, ATM, Fiber Cable, etc.
Layer 1	Physical	RS232, 100BaseTX, ISDN, 11.

Protocols at Application layer

- **TELNET:** Telnet stands for the **TEL**etype **NET**work. It helps in terminal emulation. allows Telnet clients to access the resources of the Telnet server. It is used for managin files on the internet. It is used for the initial setup of devices like switches. The teln command is a command that uses the Telnet protocol to communicate with a remote devic or system. Port number of telnet is 23.
- **FTP:** FTP stands for file transfer protocol. It is the protocol that actually lets us transf files. It can facilitate this between any two machines using it. But FTP is not just a protoc but it is also a program. FTP promotes sharing of files via remote computers with reliab and efficient data transfer. The Port number for FTP is 20 for data and 21 for control.
- **TFTP:** The Trivial File Transfer Protocol (TFTP) is the stripped-down, stock version FTP, but it's the protocol of choice if you know exactly what you want and where to find It's a technology for transferring files between network devices and is a simplified versio of FTP. The Port number for TFTP is 69.

Continued….

- **SMTP:** It stands for Simple Mail Transfer Protocol. It is a part of the TCP/IP protocol. Using a process called “store and forward,” SMTP moves your email on and across networks. It works closely with something called the Mail Transfer Agent (MTA) to send your communication to the right computer and email inbox. The Port number for SMTP is 25.
- **SNMP:** It stands for Simple Network Management Protocol. It gathers data by polling the devices on the network from a management station at fixed or random intervals, requiring them to disclose certain information. It is a way that servers can share information about their current state, and also a channel through which an administrate can modify pre-defined values. The Port number of SNMP is 161(TCP) and 162(UDP).
- **DNS:** It stands for Domain Name System. Every time you use a domain name, therefore, a DNS service must translate the name into the corresponding IP address. For example, the domain name www.abc.com might translate to 198.105.232.4. The Port number for DNS is 53

Continued…

- **DHCP:** It stands for Dynamic Host Configuration Protocol (DHCP). It gives IP addresses to hosts. There is a lot of information a DHCP server can provide to a host when the host is registering for an IP address with the DHCP server. Port number for DHCP is 67, 68.

Presentation layer protocols

- **MPEG**: **The Moving Pictures Experts Group's standard for the compression and coding of motion video for CD's** is very popular. QuickTime: This is for use with Machintosh or Power PC programs, it manages audio and video applications.
- **SSL (Secure Socket Layer)** and **TLS** (Transport Layer Security) are popular cryptographic protocols that are used to imbue web communications with integrity, security, and resilience against unauthorized tampering.

Session layer protocols

- **NetBIOS** is a **non-routable OSI Session Layer 5 Protocol** and a service that allows applications on computers to communicate with one another over a local area network (LAN). NetBIOS was developed in 1983 by Sytek Inc. as an API for software communication over IBM PC Network LAN technology.
- **SAP**: The protocol used by SAP programs that communicate using the NI interface is called the SAP Protocol. This is an enhanced version of the **TCP/IP protocol**, which has been supplemented by one length field and some options for error information .

Transport layer protocol

- **Transmission Control Protocol (TCP) -** is a transport protocol that **is used on top of IP to ensure reliable transmission of packets**. TCP includes mechanisms to solve many of the problems that arise from packet-based messaging, such as lost packets, out of order packets, duplicate packets, and corrupted packets.
- User Datagram Protocol (UDP) – **a communications protocol that facilitates the exchange of messages between computing devices in a network**. It's an alternative to the transmission control protocol (TCP). In a network that uses the Internet Protocol (IP), it is sometimes referred to as UDP/IP.

Internet protocol

- **IPV6**: Internet Protocol version 6 is the most recent version of the Internet Protocol, the communications protocol that provides an identification and location system for computers on networks and routes traffic across the Internet.
- **ICMP:** ICMP is a network level protocol. ICMP messages **communicate information about network connectivity issues back to the source of the compromised transmission**. It sends control messages such as destination network unreachable, source route failed, and source quench.
- **MPLS: Multiprotocol Label Switching**, or MPLS, is a networking technology that routes traffic using the shortest path based on "labels," rather than network addresses, to handle forwarding over private wide area networks.
- **ARP:** ARP is **the protocol used to associate the IP address to a MAC address**. When a host wants to send a packet to another host, say IP address 10.5. 5.1, on its local area network (LAN), it first sends out (broadcasts) an ARP packet.

Data Link Layer Protocols

- **PPP**: In computer networking, Point-to-Point Protocol is a data link layer communication protocol between two routers directly without any host or any other networking in between. It can provide connection authentication, transmission encryption, and data compression.
- **ATM:** ATM is a core protocol used in **the SONET/SDH backbone of the public switched telephone network (PSTN)** and in the Integrated Services Digital Network (ISDN), but has largely been superseded in favor of next-generation networks based on Internet Protocol (IP) technology

Physical layer Protocols

- **ISDN:** ISDN or Integrated Services Digital Network is a **circuit-switched telephone network system** that transmits both data and voice over a digital line. You can also think of it as a set of communication standards to transmit data, voice, and signaling. These digital lines could be copper lines.
- **100Base-TX:** 100Base-TX is **an Ethernet networking standard (IEEE 802.3u standard.)** that supports up to 100 Mbps transfer speed. 100Base-TX was also called as FastEthernet, because Ethernet was 10 Mbps that time and FastEthernet was faster than Ethernet.

Example OSI as a **courier service story**:

Application = Customer

Transport = Courier service

Network = Route planner

Data Link = Local delivery

Physical = Road

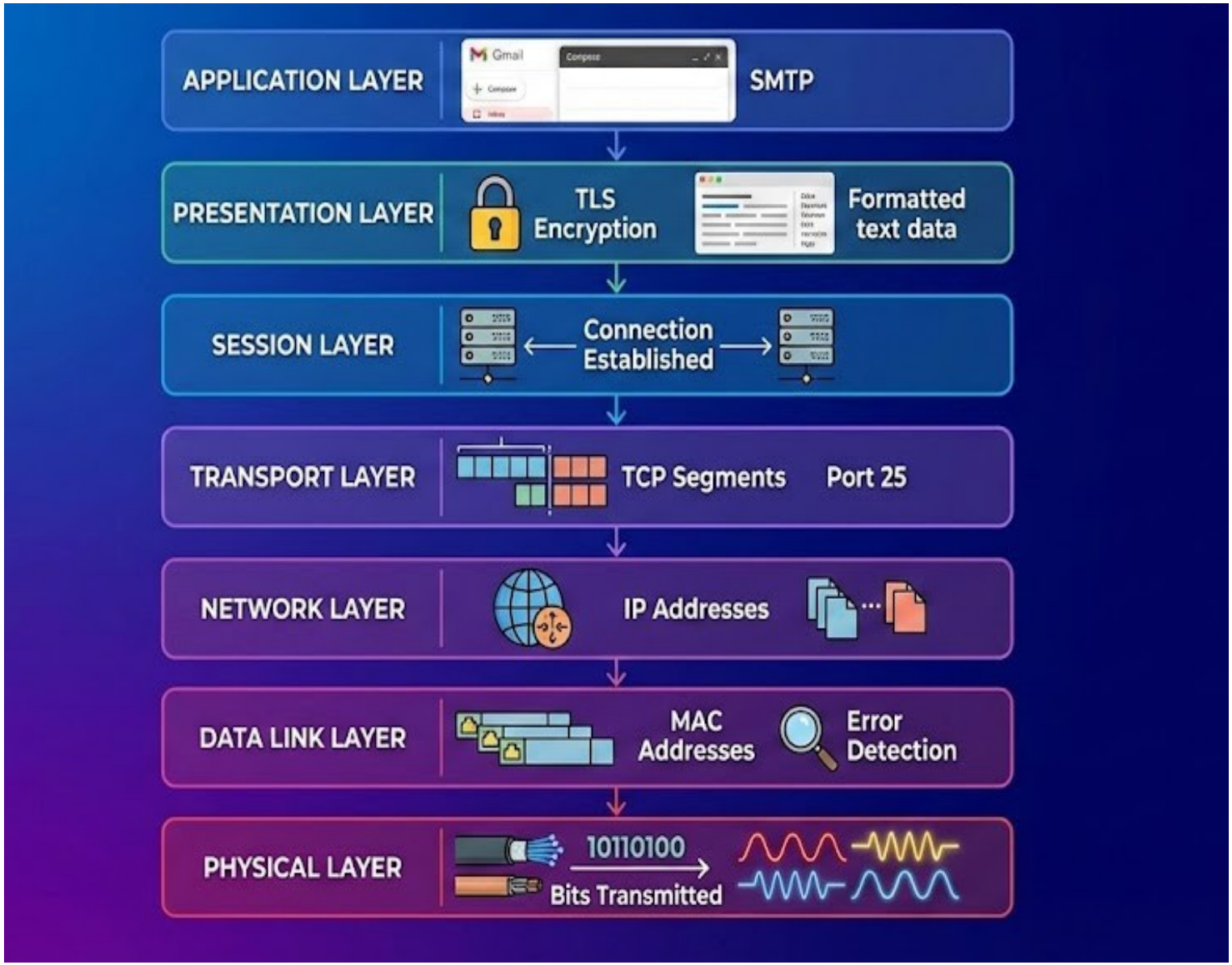

Figure 19 Email Transmission Data Flow in OSI Model

Layer #	Layer Name	Primary Function	Unit of Data (PDU)	Common Protocols & Devices	Addressing / Identifier
7	Application	Network process to application. It provides services directly to user applications (like a web browser).	Data	HTTP, HTTPS, FTP, SMTP, DNS, SSH	Specialized User IDs(e.g., URL, Email Address)
6	Presentation	Data representation and encryption. It translates data between the application and network formats (syntax layer).	Data	SSL/TLS, JPEG, GIF, ASCII, EBCDIC	N/A
5	Session	Interhost communication. It establishes, manages, and terminates connections between applications.	Data	NetBIOS, RPC, PPTP	Session ID
4	Transport	End-to-end connections and reliability. It ensures complete data transfer and error recovery.	Segment / Datagram	TCP, UDP, Firewalls	Port Numbers (e.g., Port 80, 443)
3	Network	Path determination and logical addressing (IP). It routes data packets between different networks.	Packet	IP (IPv4, IPv6), ICMP, IPSec, Routers	Logical Address (IP)(e.g., 192.168.1.5)
2	Data Link	Physical addressing (MAC). It handles error detection and data transfer between adjacent nodes.	Frame	Ethernet, Wi-Fi (802.11), PPP, Switches, Bridges	Physical Address (MAC)(e.g., 00:1A:2B:3C:4D:5E)
1	Physical	Media, signal, and binary transmission. It defines the physical specifications (volts, cables, pins).	Bit	Cables (Cat6, Fiber), Hubs, Repeaters, Network Cards	N/A

TCP/IP Protocol Suite

- The layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application.
- However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.

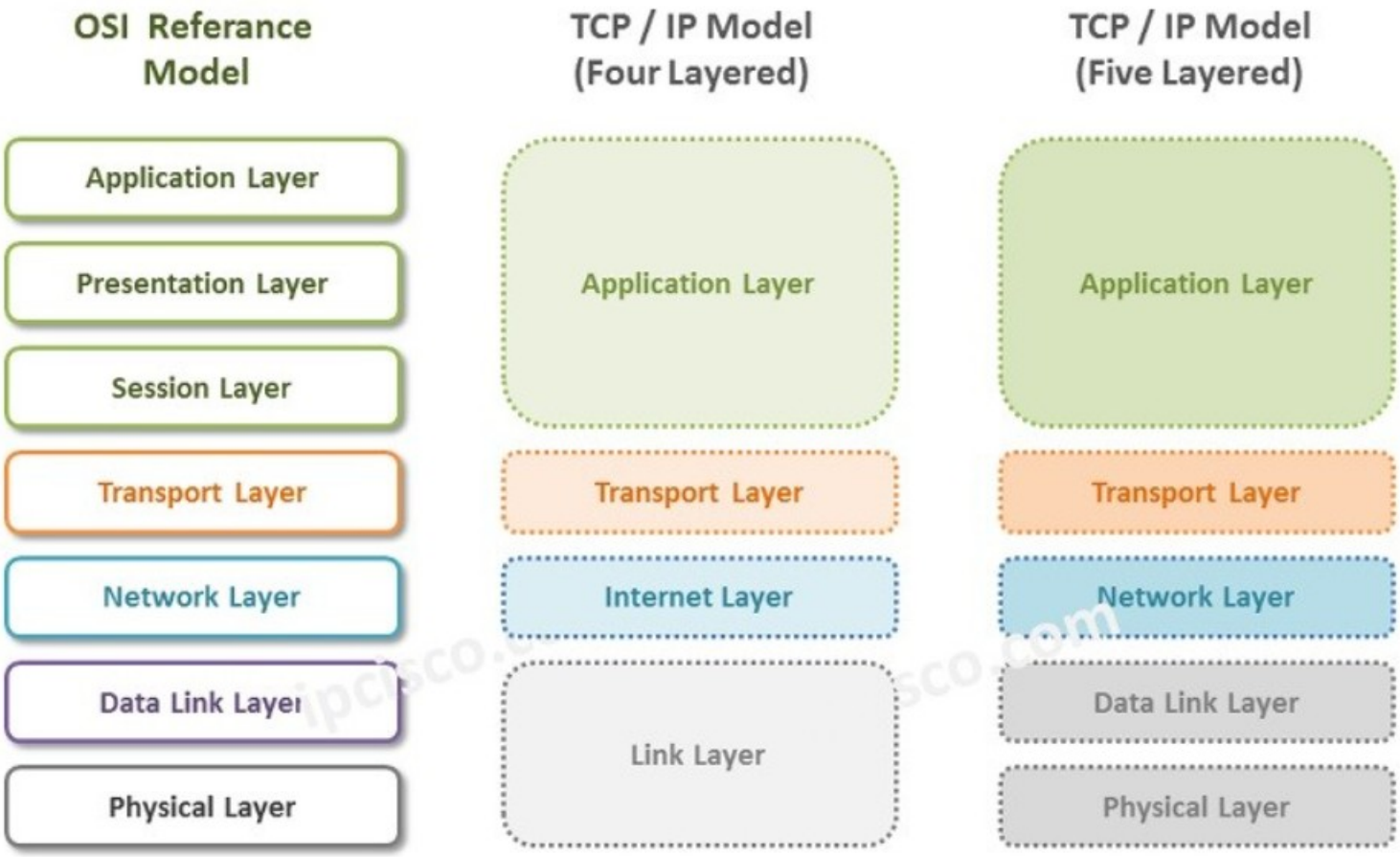

Figure 20 OSI and TCP/IP model layered structure

TCP/IP model

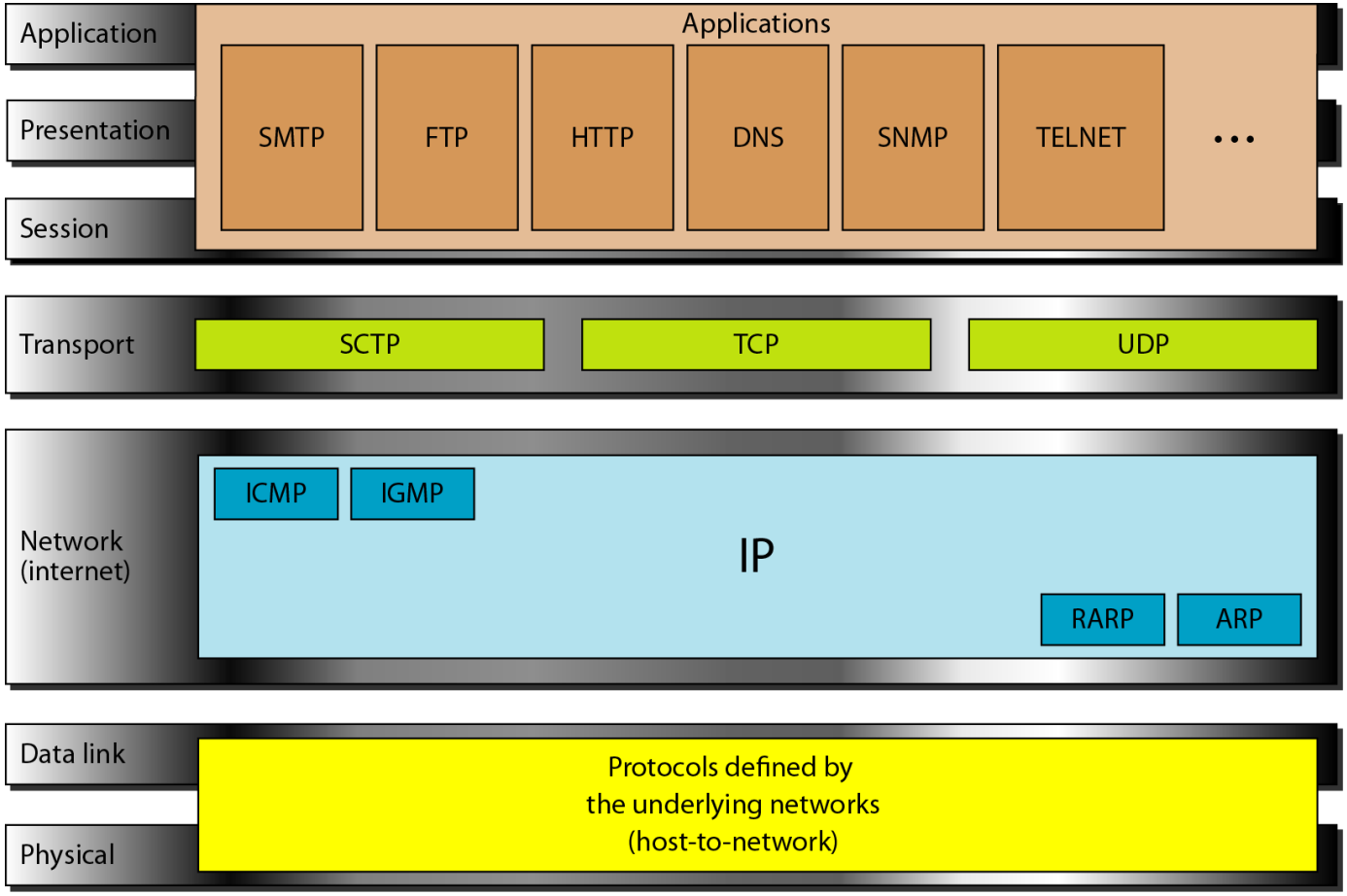

Figure 21 TCP/IP model with protocol of each layer

TCP/IP Protocol suite

Today, the TCP/IP protocol suite includes many protocols and continues to evolve to support new services. Some of the more popular ones are shown in the figure. TCP/IP is a protocol suite used by the internet and the networks of today. TCP/IP has two important aspects for vendors and manufacturers

Open standard protocol suite- Freely available to the public.
Standard-based protocols suite: It is endorsed by the networking industry and approved by standard organizations. This ensures products from different manufacturers can interoperate successfully.

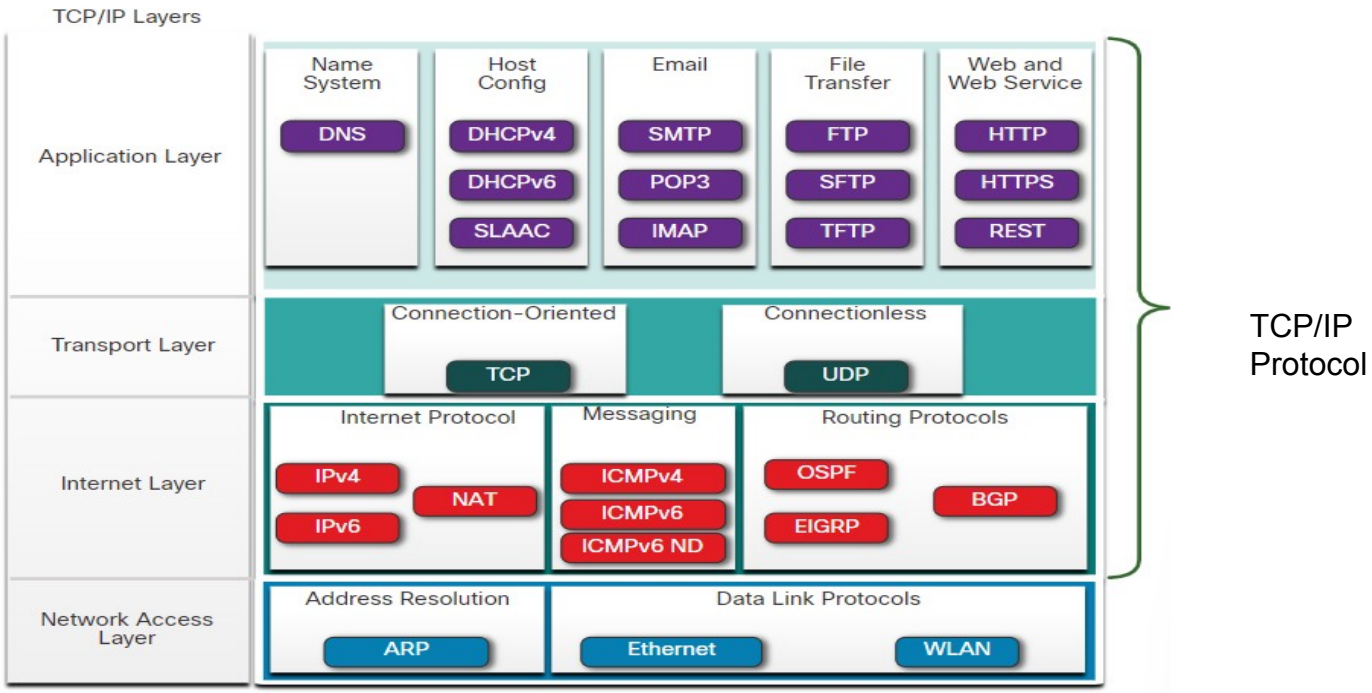

Figure 22 Shows TCP/IP protocol Suite with some popular protocol

TCP/IP Model

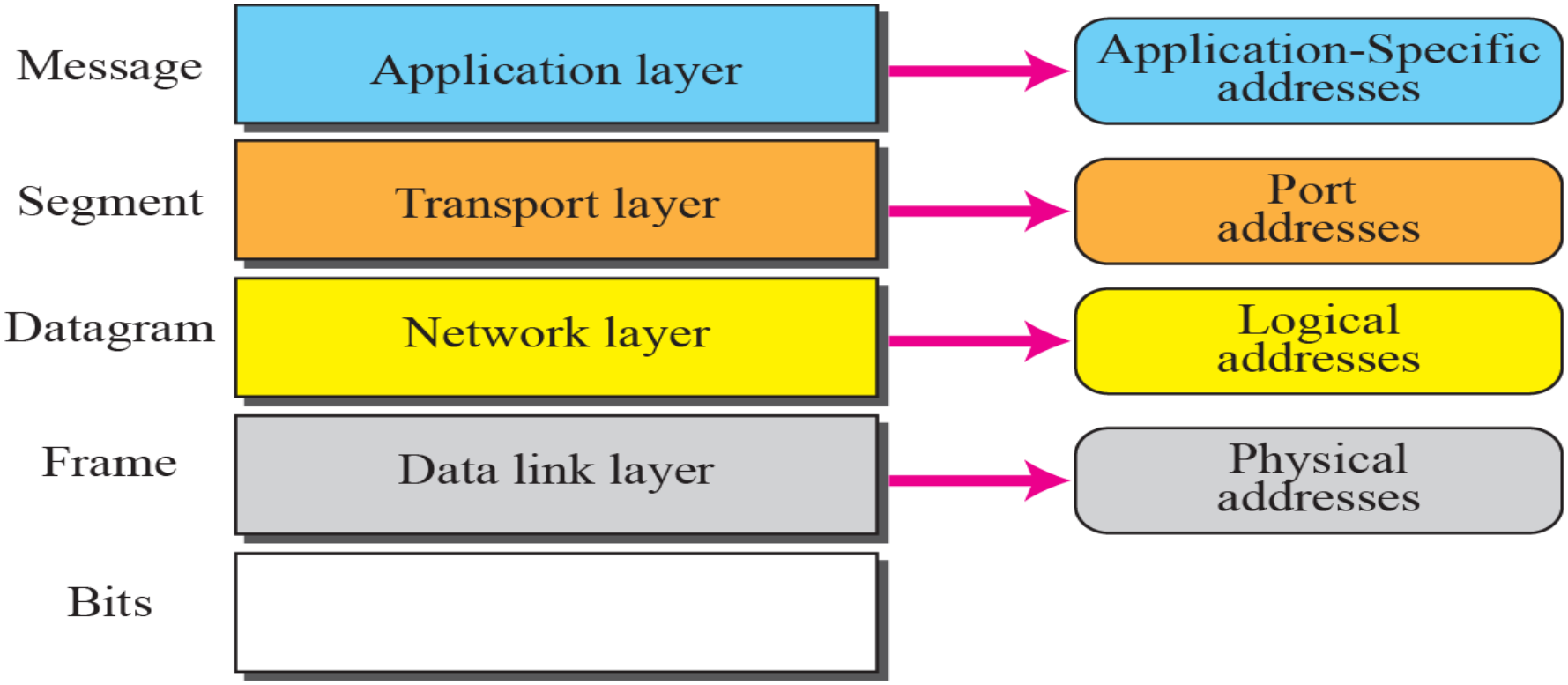

Figure 23 TCP/IP model addresses overview

TCP/IP Protocol Example

The figure 4 shows an example of the three TCP/IP protocols used to send packets between the web browser of a host and the web server. HTTP, TCP, and IP are the TCP/IP protocols used. At the network access layer, Ethernet is used in the example. However, this could also be a wireless standard such as WLAN or cellular service.

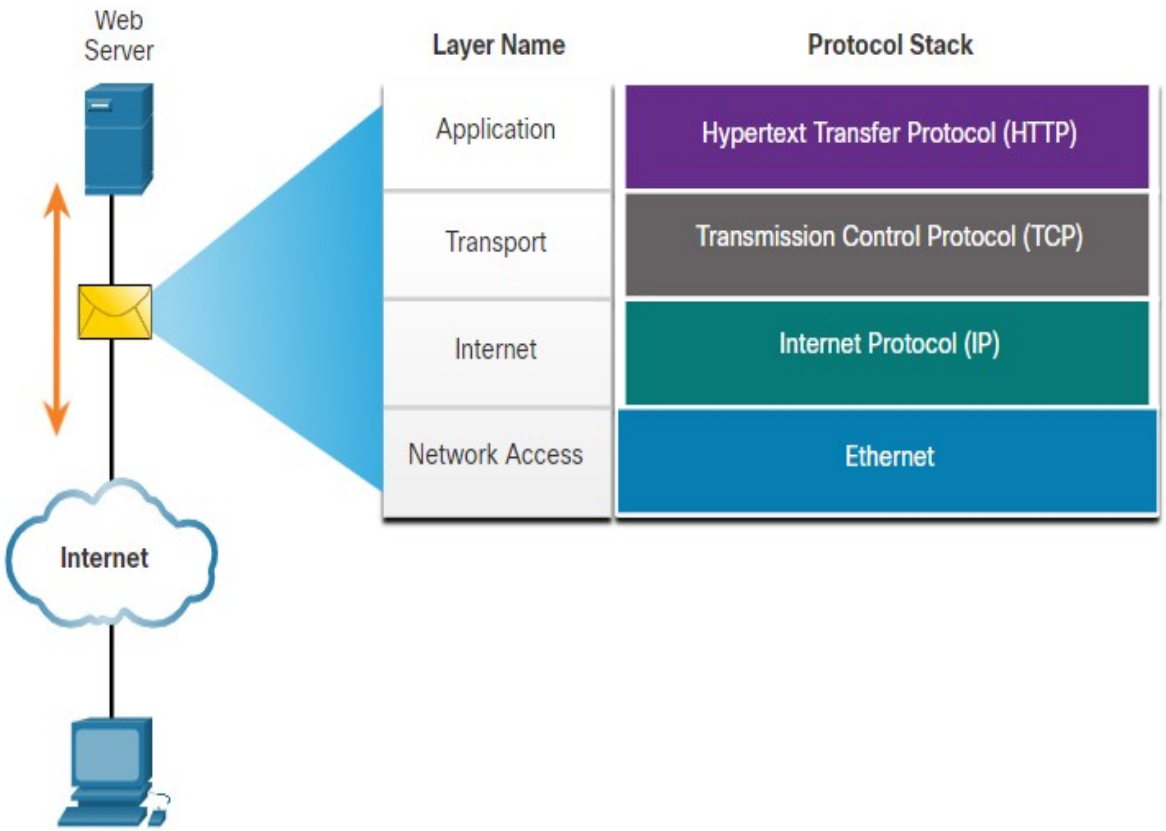

Figure 24 TCP/IP protocol example.

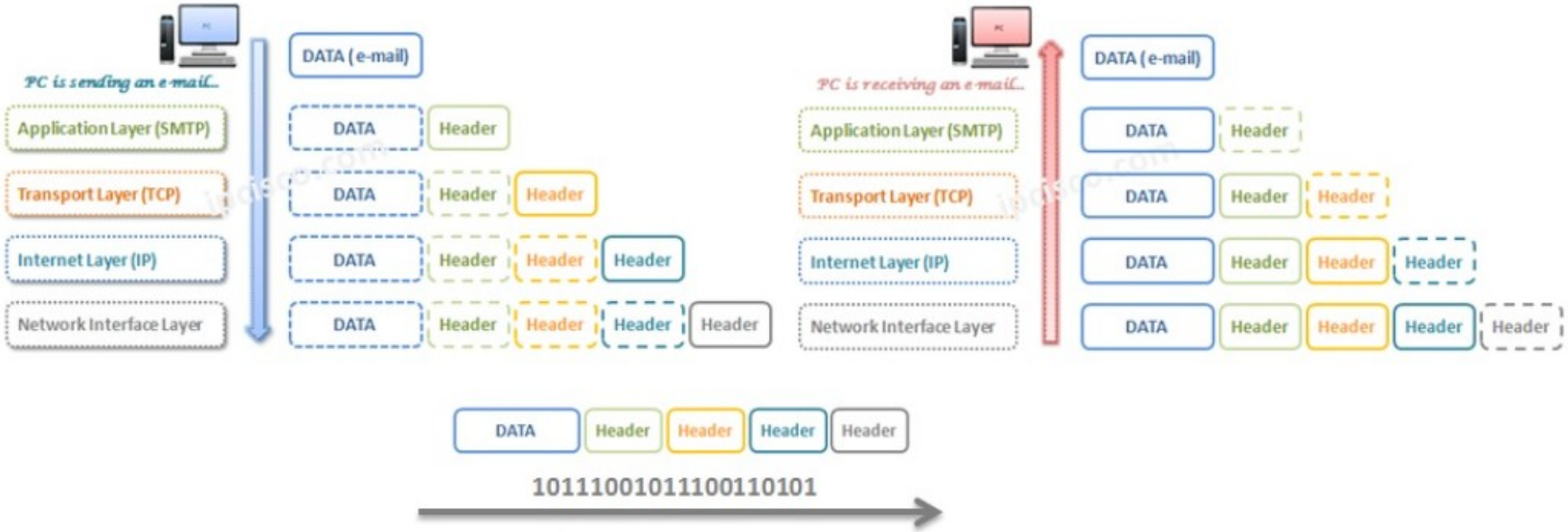

Figure 25 Exchange of information using TCP/IP model

Comparison of OSI and TCP/IP Model

- One of the major difference is that OSI is a conceptual model which is not practically used for communication, whereas, TCP/IP is used for establishing a connection and communicating through the network.

- The OSI model mainly emphasis on the services, interfaces and protocols; make a clear distinction between these concepts. Conversely, the TCP model is not able to distinctly describe these concepts.

- Furthermore, the TCP/IP enables only connectionless communication mode in the network layer but both modes (Connectionless and connection-oriented) in the transport layer. When it comes to the OSI model, it supports connectionless and connection-oriented communication over the network layer but in the transport layer, connection-oriented communication is merely allowed.

Comparison of OSI and TCP/IP Model

- OSI (Open System Interconnection) model was introduced by ISO (International Standard Organization). Currently OSI (Open System Interconnection) model is just a reference model. Means that, there are no real working implementations of OSI Model on any of latest network infrastructure devices or Operating Systems. OSI (Open System Interconnection) model is used these days only as a reference model for teaching computer networking concepts and for understanding how computer networks operate.

- The roots of TCP/IP (Transmission Control Protocol Internet Protocol) are with US Department of Defense (DoD). Original TCP/IP (Transmission Control Protocol/Internet Protocol) model had only four layers, and later an updated model evolved with five layers. TCP/IP protocol suite and thus TCP/IP model got wider acceptance than OSI model, because world's largest network internet started operating using TCP/IP protocol suite.

Comparison of OSI and TCP/IP Model

- TCP/IP model is developed by Department of Defence, whereas OSI model was developed by ISO.
- Four layers in TCP/IP whereas seven layers in OSI model.

TCP/IP	OSI
TCP refers to Transmission Control Protocol.	OSI refers to Open Systems Interconnection.
TCP/IP has 4 layers.	OSI has 7 layers.
TCP/IP is more reliable	OSI is less reliable
TCP/IP does not have very strict boundaries.	OSI has strict boundaries
TCP/IP follow a horizontal approach.	OSI follows a vertical approach.
TCP/IP uses both session and presentation layer in the application layer itself.	OSI uses different session and presentation layers.
TCP/IP developed protocols then model.	OSI developed model then protocol.
Transport layer in TCP/IP does not provide assurance delivery of packets.	In OSI model, transport layer provides assurance delivery of packets.
TCP/IP model network layer only provides connection less services.	Connection less and connection oriented both services are provided by network layer in OSI model.
Protocols cannot be replaced easily in TCP/IP model.	While in OSI model, Protocols are better covered and is easy to replace with the change in technology.

Physical Address (MAC Address)

- **Layer:** Data Link Layer (Layer 2).
- **Definition:** A 48-bit unique identifier burned into the Network Interface Card (NIC) by the manufacturer.
- **Purpose:** Local delivery between directly connected nodes on the same network (subnet).
- **Scope:** Changes at each hop (router) along the path from source to destination.
- **Example:** 00:1A:2B:3C:4D:5E.

Logical Address (IP Address)

- **Layer:** Network Layer (Layer 3).
- **Definition:** A hierarchical address (32-bit for IPv4, 128-bit for IPv6) that represents a host's location in a network.
- **Purpose:** Routing data packets across different networks (globally).
- **Scope:** Remains constant from source to final destination.
- **Example:** 192.168.1.1.

Port Address (Port Number)

- **Layer:** Transport Layer (Layer 4).
- **Definition:** A 16-bit identifier (0 to 65,535) assigned by the OS to specific applications or services.
- **Purpose:** Identifies specific processes or applications (e.g., Web, Email) on a host.
- **Example:** Port 80 (HTTP), Port 443 (HTTPS).

Summary of Differences

Feature	Physical Address (MAC)	Logical Address (IP)	Port Address
Layer	Data Link (2)	Network (3)	Transport (4)
Scope	Local (Same Network)	Global (Across Networks)	Process/Application
Uniqueness	Globally Unique	Unique per Network	Unique per Host
Change	Changes per Hop	Constant	Constant
Identifies	NIC/Hardware	Host/Device	Application/Process

Internet not working but Wi-Fi connected decide **which OSI layer to check first** and *why*

NETWORK LAYER (Layer 3

Because **Wi-Fi connected** already tells us:

- **Physical layer (L1)** is working
- **Data Link layer (L2)** is working (Wi-Fi association, MAC address OK)

If the internet is not working, the **most common issue is at the Network layer**, such as:

•No **valid IP address**
•Wrong **default gateway**
•Routing problem
•DHCP failure